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#include "mcc_generated_files/system/system.h"
#include <xc.h>

// SIMPLE SIREN USING DAC1

int8_t direction;
uint8_t dacValue=0;
int8_t stepSize=5 ;

void Siren_Update(void)
{
    // direction = 1 → HIGH PITCH
    // direction = 0 → LOW PITCH

    if (direction == 1)
    {
        dacValue = 100    ;
        direction = 0;
    }
    else
    {
        dacValue = 10;
        direction = 1;
    }

    DAC1_SetOutput(dacValue);
}

int main(void)
{
    SYSTEM_Initialize();

    DOOR_SetDigitalInput();    DOOR_SetDigitalMode();
    SWITCH_SetDigitalInput();  SWITCH_SetDigitalMode();
    LED_SetDigitalOutput();    LED_SetDigitalMode();
    MOTOR_SetDigitalOutput();  MOTOR_SetDigitalMode();
    SPEAKER_SetDigitalOutput();
    SPEAKER_SetDigitalMode();

    uint8_t override = 0;
    uint8_t lastSwitch = SWITCH_GetValue();

    uint8_t lastDoor = DOOR_GetValue();    // track door open or close

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while(1)
{
    // SWITCH TOGGLE DETECTION

    uint8_t now = SWITCH_GetValue();

    if (now != lastSwitch)
    {
        __delay_ms(20); // debounce

        if (now == SWITCH_GetValue())
        {
            if (now == 1)
            {
                override ^= 1; // toggle state

                LED_SetLow();
                DAC1_SetOutput(0);
            }
        }
    }

    lastSwitch = now;

    // MOTOR OUTPUT BASED ON SWITCH STATE

    if (override == 1)
        MOTOR_SetHigh(); // override ON = MOTOR SIGNAL SENDS HIGH
    else
        MOTOR_SetLow(); // override OFF = MOTOR SIGNAL SENDS LOW

    // OVERRIDE ENABLED = EVERYTHING OFF
    if (override == 1)
    {
        LED_SetLow();
        DAC1_SetOutput(0);
        continue;
    }

    // DOOR CLOSED = EVERYTHING OFF

    uint8_t doorNow = DOOR_GetValue();

```

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if (doorNow == 0)
{
    LED_SetLow();
    DAC1_SetOutput(0);
    lastDoor = 0;
    continue;
}

// DOOR JUST OPENED = SIREN START

if (lastDoor == 0 && doorNow == 1)
{
    DAC1_SetOutput(0);
}

lastDoor = 1;

// DOOR OPEN = LED ON + UPDATE DAC SIREN

LED_SetHigh();

// Update DAC waveform every loop
Siren_Update();

__delay_us(10 );
}
}

```